An AI-based Approach to Intelligent Waste Collection Optimization

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WASTE COLLECTOR

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PROBLEM STATEMENT

Introduction: Problem to be Solved

Current Inefficiency

Traditional collection relies on fixed schedules determined by historical patterns rather than actual service demand.

Service Failure

Fixed schedules lead to overflows, resulting in street litter, odors, and degraded public hygiene.

Operational Waste

Trucks often visit empty bins, resulting in unnecessary fuel consumption and operational expenses.

🚩Goal

Implement demand-driven strategies using real-time IoT sensor data.

BACKGROUND

Background & State of the Art

1

Static Schedules

Traditional fixed-route collection without real-time data

2

Threshold-based (Reactive)

Collection triggered when bins reach capacity thresholds

3

AI Forecasting (Proactive)

Predictive models anticipate fill levels for optimized scheduling

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Context

Recent deployment of IoT sensors enables continuous real-time monitoring of fill levels.

Limitations of Heuristics

Naive threshold-based strategies fail to exploit predictive information for efficient scheduling.

Literature Foundation

Deep Learning: Ahmed et al. (2022) showed LSTM efficacy reporting R^2 values around 0.93

Hybrid Models: Prophet captures multi-seasonal patterns

Exogenous Features: Weather, location metadata, holidays, etc. can influence human behaviour

RESULTS

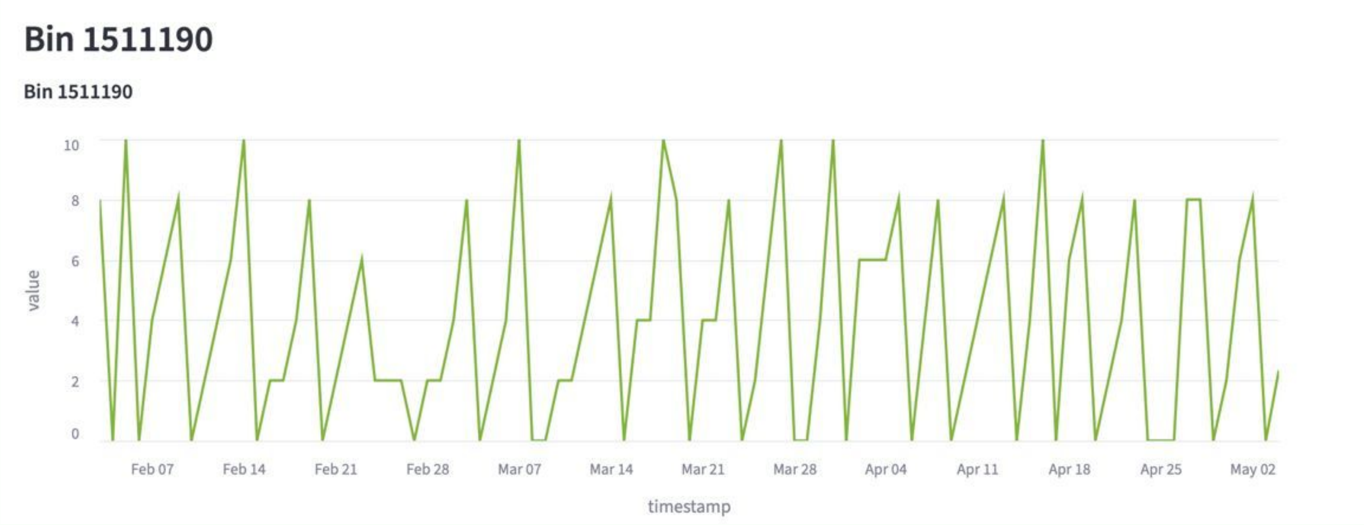
Proposed Solution:

Prior work

- Prior literature used deep learning on the Melbourne Garbage Collection Dataset
- Dataset: daily fill-levels of smart garbage bins

Augmentation (Exogenous features)

- Sun hours
- Rain (L/m^2)
- Commercial pressure (location meta data)
    - (commercial buildings / bins within a radius)
- Weather forecast (next 2 days)



Preprocessing

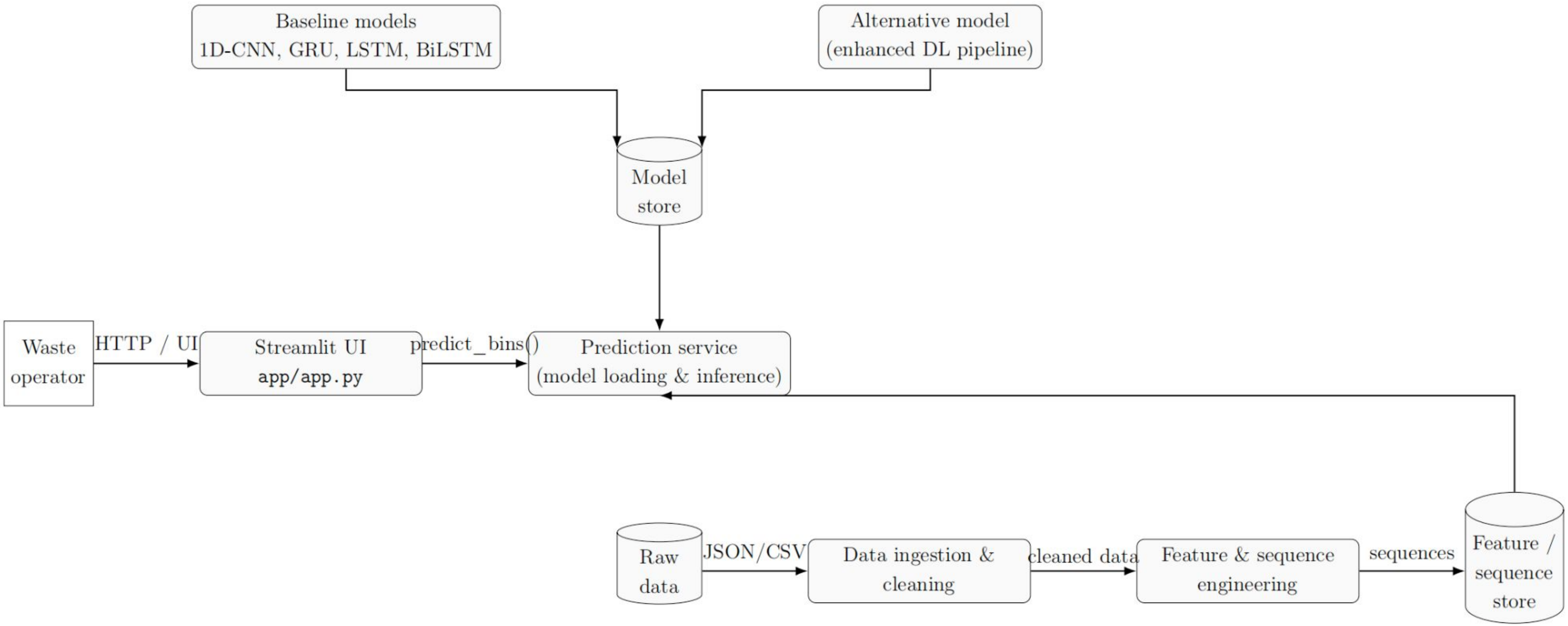
- Remove collection effects from the learning signal
- Don't use absolute fill-levels
- Use daily fill rates:
    - "How much was added to the bin today?"

Model

- LSTM
- 30-day input sequence (window length = 30)

ARCHITECTURE

Proposed Solution: Functional Architecture



Proposed Solution: System Components

Dataset

Melbourne Argyle Square

- 32 Smart Bins monitored
- Period: July 2018 - May 2021
- Daily fill-level measurements

Models

Baseline: 1D-CNN, GRU, LSTM, BiLSTM

Alternatives: Prophet, NeuralProphet, XGBoost (with exogenous features), LSTM (with exogenous features)

UI

Interactive dashboard (Streamlit)

- Bin selection
- Parameter configuration
- Forecasting visualization
- CSV export capability


Fill Levels
Fill Rates
Controls
Amount of days to predict
7
Selection method:
Dropdown
Map-based
Select bin serialNumber(s)
1510830
1511190
1511191
1511192
Predict
Deploy
Waste Collector — Fill-rate (Accumulated)
WASTE COLLECTOR
Predicted cummulative Fill-Rates
Fill level
0
2
4
6
8
10
12
14
16
18
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Weekday / Date
1510830
1511190
1511191
1511192
1511198
1511199
1511200

RESULTS

Results: Performance & Explainability

Accuracy

Machine learning models achieves moderate accuracy capturing the main fill-level trends, useful for scheduling.

Transparency

Integrated SHAP (SHapley Additive exPlanations) analysis reveals the contribution of features like holidays and weather.

Model Comparison

Alternative models with enriched features (weather, calendar attributes, location metadata) improved upon baselines.

Validation

Reproduced literature baselines (LSTM/GRU) effectively captured temporal patterns.

Results: Accomplishments

1

End-to-End Pipeline

From raw data ingestion to user-facing prediction service.

3

Reproducibility

Complete GitHub repository with code, seeds, and documentation.

2

Functional Prototype

Deployed Streamlit application enabling interactive forecast generation.

4

Sustainability Focus

Integrated economic cost-benefit and environmental impact assessments.

ECONOMICS

Economic Analysis

€76,052

Investment (CAPEX)

Total development cost

Human Resources: €75,172 (Team of 5 FTEs).

Technical Infrastructure: €880 (GPU compute & storage).

€36,308

Annual Operational Savings

Fuel savings: ~€12,276 annually (14,400 km reduction)

Maintenance & tire savings: ~ €4,032 annually

Labor efficiency: ~€20,000 annually

2.1

Return on Investment

Payback Period: Approximately 2.1 years

Long-term Strategy: Operational lifetime expected to exceed 5 years.

Cost Comparison(5-Year Horizon)

Commercial SaaS: ~€450,000+ (€15-20/bin/month).

In-House Solution: ~€86,000 (Initial CAPEX + Maintenance).

Conclusion: >80% cost reduction compared to commercial alternatives.

Sustainability Analysis

Environmental

Carbon Footprint: Direct reduction of 15 - 22 tonnes of CO_2 annually.

Life Cycle Benefit: Up to 19 - 28 tonnes CO_2 when including upstream fuel production savings.

Equivalence: Offsets the annual emissions of 3 to 5 passenger cars.

Air Quality: Measurable decrease in NOx and Particulate Matter in high-traffic urban corridors.

Social

Service Quality: 40 - 60% reduction in bin overflow events.

Public Health: Reduced pest activity (rodents/insects) and improved street hygiene.

Equity: Monitoring ensures service quality is maintained across diverse neighborhoods.

Governance & Response AI

Regulatory Alignment: Designed to anticipate the EU Artificial Intelligence Act.

Transparency: SHAP analysis ensures decisions are explainable, preventing 'black box' risks.

Data Privacy: Governance frameworks protect behavioral patterns inferred from waste generation.

MANAGEMENT

Project Management: Methodology

Agile Methodology

- **Framework:**
  Agile practices were adapted to the student context, but uneven academic schedules made fully synchronized sprints more difficult.
- **Planning:**
  Weekly meetings and monthly reviews ensured coordination despite these constraints.

Collaboration & Management Tools

- **Version Control:** GitHub central repository for collaborative development.
- **Task Tracking:** shared Kanban board to visualize tasks and manage workload distribution.
- **Documentation:** On shared file

Technical Workflow Git Strategy

- **Main:** Stable, production-ready code for deployment.
- **Baseline Approach:**
  Dedicated branch for reproducing Deep Learning paper baselines (CNN, LSTM).
- **Alternative Approach:**
  Branch for experimenting with enhanced models and feature engineering.

Project Management: Schedule

Duration: 17 Weeks (September 22, 2025 - Mid-January 2026)

Weeks 1-2 — 1
Literature Review & Baseline

2 — **Weeks 3-4**
EDA & Data Cleaning

Weeks 5-6 — 3
Model Development

4 — **Weeks 7-10**
Evaluation & UI

Weeks 16-17 — 5
Final Delivery

TEAM

Project Management: Team Assignment

Mattéo Audigier
Project Coordinator & Baseline Deep Learning Pipelines (1D-CNN, GRU)

Michał Binda
Preprocessing Pipeline & LSTM Implementation, UI Development

Jakob Leitner
Alternative Models, Feature Engineering, & Visualization

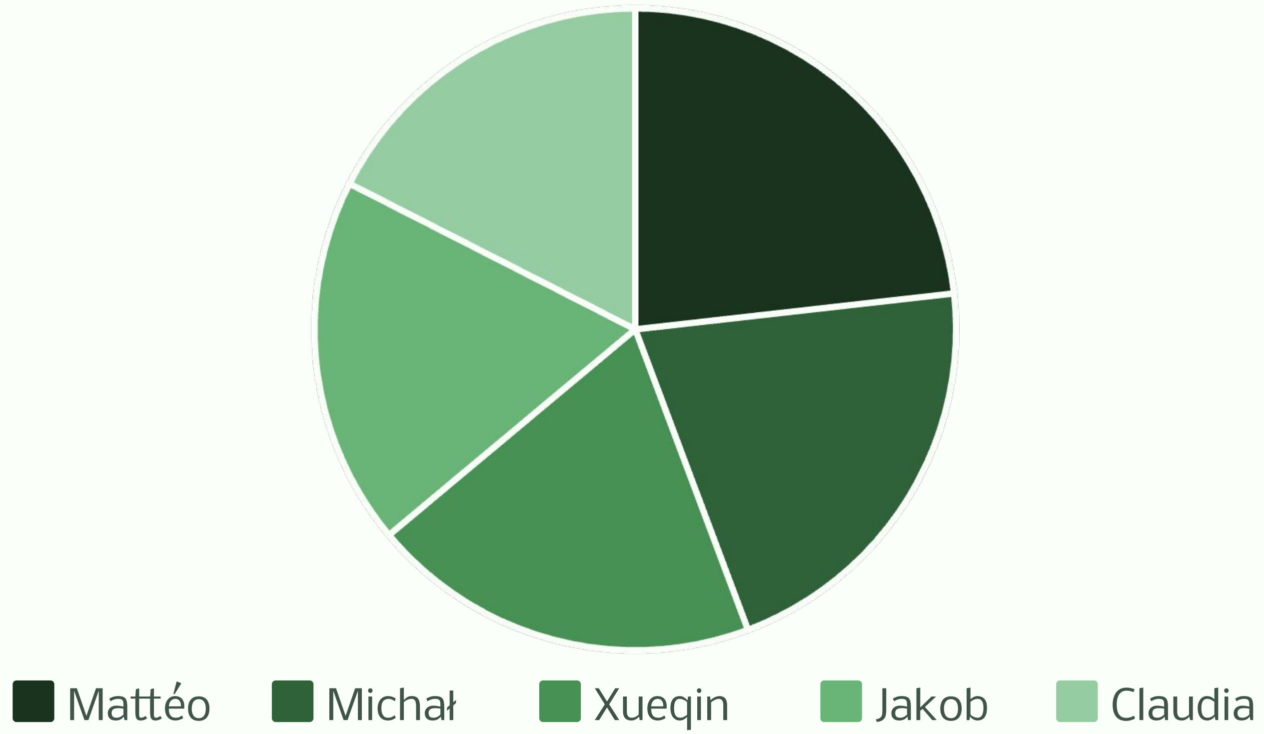
Claudia De Carlo
Sustainability Lead, Economic Analysis

Xueqin Tian
Documentation Lead, User Manual, & Economic Estimation

Project Management: Effort Distribution

Total Effort

Approximately **348 hours** invested across all team members.



Individual Contributions

Mattéo Audigier: 77h

Michał Binda: 73h

Xueqin Tian: 69h

Jakob Leitner: 66h

Claudia De Carlo: 63h

Intensive Phase

Significant effort during the final delivery phase (Weeks 16-17).

CONCLUSIONS

Conclusions

Success
Validated technical feasibility of AI forecasting for waste optimization.

Value
Measurable economic and environmental benefits justify the technology.

Deliverables
Complete, reproducible system architecture and documentation.

Limitations
Dataset limited to Melbourne (generalization constraint) and lack of vehicle routing integration.

NEXT STEPS

Future Work

Route Optimization
Integrate algorithms to automate daily collection schedules based on forecasts.

Real-Time Data
Transition to continuous data ingestion and automated forecast updates.

Transfer Learning
Adapt models to new geographic contexts (e.g., European cities).

Uncertainty Quantification
Implement prediction intervals to better manage overflow risks.

Thank You!

Project Team:

Mattéo Audigier,

Jakob Leitner,

Michał Binda,

Claudia De Carlo,

Xueqin Tian

Questions?